Knowing When Not to Answer: Noise-Conditional Hallucination Risk Estimation and Shift-Robust Selective Generation for Retrieval-Augmented Language Models

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Abstract

Retrieval-augmented generation (RAG) is widely deployed to reduce hallucination in large language models, yet retrieval is itself a noise source: irrelevant, contradictory, or misleading evidence can induce hallucinations a model would not otherwise produce. We study selective generation for RAG, where a system answers only when its hallucination risk is statistically controlled, under the realistic condition that the retrieval-noise distribution shifts between calibration and deployment. Using a controlled two-regime testbed that separates parametric from retrieval-induced error, we establish three behavioural regularities: a single unopposed false document is absorbed at 85 to 100 percent across three generators of two architectures, even against facts a model knows (context dominance); contradictions at the first context position are more than twice as damaging as at the last (misinformation primacy); and under conflicting evidence all errors are confident assertions rather than abstentions (silent failure). We then show that a noise-conditional risk estimator using observable retrieval descriptors predicts hallucination far better than model-internal uncertainty, beating sampling-based semantic entropy by 0.31 to 0.40 AUROC across six instruction-tuned model families. Standard conformal risk control and its covariate-shift remedies all violate a 10 percent target in 200 of 200 resplits under noise shift; we introduce risk-binned Mondrian conformal risk control, prove its validity for bin-measurable shifts, and show it eliminates violations at competitive answer rates, with a label-free deployment monitor that makes its assumption checkable. Results hold across eight generators from 80M to 9B parameters, five model families, two architectures, and two content domains. Code and data are released.

1 Introduction

Retrieval-augmented generation [Lewis et al., 2020, Gao et al., 2023] grounds large language models (LLMs) in external corpora and is routinely presented as a mitigation for hallucination [Huang et al., 2025, Zhang and Zhang, 2025]. The argument rests on an assumption that deployment rarely satisfies: that retrieved evidence is relevant and correct. Production retrieval pipelines return passages that are off-topic, partially relevant, stale, about confusable entities, or factually wrong [Chen et al., 2024, Wang et al., 2025, Cuconasu et al., 2024]. Worse, the knowledge corpus

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itself can be corrupted: Zou et al. [2025] show that injecting a handful of crafted documents reliably steers RAG answers, and defenses remain weak. A growing literature documents that LLMs over-trust retrieved context even against their own parametric knowledge [Xie et al., 2024, Kortukov et al., 2024, Xu et al., 2024], so that RAG can paradoxically suppress appropriate abstention because added context raises confidence even when that context is wrong. When evidence is bad, RAG does not merely fail to help; it manufactures fluent, confident errors.

For high-stakes applications the question is therefore not only how to make RAG more robust, but when a RAG system should decline to answer. Selective prediction has a long history in classification [Chow, 1970, Geifman and El-Yaniv, 2017]; for generation, conformal methods provide finite-sample guarantees on error rates among answered queries [Angelopoulos et al., 2024, Abbasi-Yadkori et al., 2024, Quach et al., 2024, Mohri and Hashimoto, 2024]. Existing conformal treatments of RAG, however, either certify population-level risk [Kang et al., 2024] or assume the calibration data is exchangeable with deployment [Li et al., 2024, Feng et al., 2025]. Retrieval noise breaks exchangeability in a specific and practically common way: the mixture of noise regimes at deployment (which retriever, which corpus, which adversary) differs from the mixture at calibration. A system calibrated on clean development traffic and deployed against a degraded or poisoned corpus faces exactly this shift.

This paper studies per-decision risk control for RAG under retrieval-noise shift. We ask three questions. First, can hallucination be predicted before answering from observable signals, and how much do retrieval-side descriptors add over the model-internal uncertainty measures that dominate current practice [Kuhn et al., 2023, Farquhar et al., 2024, Kossen et al., 2024]? Second, how do noise type, severity, and position causally shape hallucination and abstention, and how do these effects interact with parametric knowledge? Third, when standard conformal calibration fails under noise shift, what restores validity, and at what cost in answer rate?

Causal answers to the second question require intervening on evidence while holding queries fixed, and require separating parametric error (the model would have been wrong anyway) from retrieval-induced error (the noise caused the mistake). Naturalistic benchmarks support neither. We therefore construct a controlled two-regime testbed. A synthetic world of 240 fictional scientists guarantees that no pretrained model holds parametric knowledge of the queried facts, so closedbook hallucination is 100% by construction and every correct answer is attributable to retrieval. Sixty well-known real facts (country capitals and currencies) provide the complementary regime in which mid-size models answer correctly without context 60% to 90% of the time. Typed noise is injected into a BM25 pipeline at graded severities and controlled context positions: partial and total irrelevance, contradictions at first and last rank, unopposed misinformation, confusable-entity distractors, and compound mixtures. Every generation is logged with retrieval-side, model-side, and closedbook-contrast features and labeled exactly against the closed answer space. The released dataset comprises 6,230 generations on the main domain across three generators spanning encoder-decoder and decoder-only architectures, plus 2,970 generations on an independent second domain.

Our contributions are as follows.

1. **Behavioural regularities of noisy RAG** (§5.1). We document three phenomena. Context dominance: one unopposed false document yields hallucination rates of 85% to 100% across all three generators, including on facts the models answer correctly without context. Misinformation primacy: a contradiction at rank 1 is more than twice as damaging as the same document at rank 4 (53% versus 23% hallucination without parametric support; 30% versus 0% with it). Universal silent failure: under every conflict-noise type, 100% of errors are confident assertions; behavioural abstention occurs only under irrelevance noise. The

negative-rejection ability targeted by current instruction tuning [Chen et al., 2024, Zhang et al., 2024] detects absent evidence but not deceptive evidence.

2. **Noise-conditional risk estimation** (§5.2). Combining retrieval descriptors, model-internal signals, and closedbook-contrast features raises hallucination-prediction AUROC from 0.788 to 0.915. A comparison against an earlier campaign with cruder noise shows the margin grows from 2.6 to 12.7 points as noise becomes subtler, inverting the natural objection that retrieval features are redundant with model confidence. Per-condition diagnostics locate the residual difficulty precisely in conflict noise (AUROC 0.60 to 0.72), the same regime in which models never warn.

3. **Shift-robust selective generation** (§5.4). Under clean-heavy calibration and noise-heavy deployment, marginal conformal risk control and its standard covariate shift remedies, learned-group calibration and density-ratio weighting [Tibshirani et al., 2019], violate a 10% risk target in every one of 200 resplits. Risk-binned Mondrian conformal risk control, which certifies score bins with Clopper-Pearson upper bounds, is proved valid for bin-measurable shifts (Proposition 1), eliminates violations entirely, answers 2.3 times more queries than a worst-group bound, and remains valid across target levels $\alpha \in \{0.10, 0.15, 0.20\}$, bin counts $B \geq 10$, and all tested shift severities.

4. **Transfer analysis** (§5.8). Risk estimators transfer across eight generators from 80M to 9B parameters and five model families with AUROC 0.87 to 0.96, yet transplanted thresholds realize 1.7 to 2.7 times the target risk. Discrimination is substantially model-general; calibration is not. The binned certification scheme supports cheap local recalibration, and we state the cross-lingual analogue of this experiment as a registered prediction for future work.

2 Related Work

2.1 Noise, conflict, and corruption in RAG

Benchmarks and analyses have established that retrieval quality governs RAG reliability. RGB isolates noise robustness and negative rejection as core abilities and finds models lacking in both [Chen et al., 2024]. RAGTruth provides word-level hallucination annotations for RAG outputs [Wu et al., 2024]. RAAT types retrieval noise and trains adapters against it [Fang et al., 2024]; Astute RAG consolidates internal and external knowledge at inference [Wang et al., 2025]. Cuconasu et al. [2024] report the counterintuitive finding that random irrelevant documents can improve RAG accuracy, while related distractors hurt; our graded-irrelevance results refine this picture (§6). On the adversarial side, PoisonedRAG demonstrates effective knowledge corruption attacks with few injected documents [Zou et al., 2025]. Position effects are documented for long contexts by Liu et al. [2024]; we show the analogous effect for misinformation specifically. Knowledge-conflict studies characterise when models prefer context over memory [Xie et al., 2024, Kortukov et al., 2024, Xu et al., 2024] and how to control the balance [Bi et al., 2025, Zhang et al., 2025]. This literature measures and manipulates behaviour; it does not predict per-query hallucination risk or provide decision-level guarantees.

2.2 Uncertainty quantification for LLMs

Semantic entropy estimates uncertainty over meanings rather than strings [Kuhn et al., 2023, Farquhar et al., 2024]; probes recover it cheaply from hidden states [Kossen et al., 2024]; kernel language entropy refines the geometry [Nikitin et al., 2024]. Self-evaluation signals include P(True) [Kadavath et al., 2022] and sampling consistency [Manakul et al., 2023]. These methods treat the generator as a closed system. None conditions on the quality or structure of retrieved evidence, and our results quantify the cost of that omission under retrieval noise.

2.3 Selective prediction and conformal guarantees

Selective classification with a reject option originates with Chow [1970] and was revived for deep networks by Geifman and El-Yaniv [2017]. Conformal risk control bounds the expectation of monotone losses [Angelopoulos et al., 2024]; conformal abstention applies this to hallucination [Abbasi-Yadkori et al., 2024]; conformal language modeling and conformal factuality give guarantees for generation [Quach et al., 2024, Mohri and Hashimoto, 2024]. For RAG, TRAQ ensembles conformal sets over retrieval and generation [Li et al., 2024]; Conformal-RAG filters sub-claims with group-conditional coverage [Feng et al., 2025]; C-RAG certifies an upper bound on population-level generation risk, including under bounded distribution drift [Kang et al., 2024]. Weighted conformal prediction corrects for known covariate shift via likelihood ratios [Tibshirani et al., 2019], adaptive conformal inference tracks unknown drift online [Gibbs and Candès, 2021], and the limits of distribution-free conditional inference are themselves characterised by Barber et al. [2021]. Our contribution sits at the intersection: we show empirically that likelihood-ratio weighting and unsupervised grouping both fail under noise-regime selection, and we give a binned certification scheme whose validity needs neither ratio estimation nor exchangeability, only that the calibrated score resolves the shift. We also show that population certificates and per-decision validity diverge sharply in this setting, which clarifies the relationship to C-RAG. The guarantees in this literature are of three kinds, and none is the one a selective-answering system needs. Conformal risk control and C-RAG bound population or marginal risk [Angelopoulos et al., 2024, Kang et al., 2024]; conditional conformal methods and Conformal-RAG bound conditional *coverage* rather than risk [Gibbs et al., 2025, Feng et al., 2025]; and covariate-shift methods restore only *marginal* coverage under a shifted test distribution [Tibshirani et al., 2019, Gibbs and Candès, 2021]. Weighted conformal risk control under covariate shift remains marginal rather than bin-conditional [Zecchin and Simeone, 2025], and exact feature-conditional coverage is impossible distribution-free [Barber et al., 2021], which is precisely why we partition the feature space into a small number of risk bins instead. Our risk-binned Mondrian conformal risk control is, to our knowledge, the only method that delivers per-query, risk-bin-conditional control of the answer-or-abstain risk that stays valid under measurable retrieval-noise covariate shift.

3 Problem Setup and Method

3.1 Setting and observable features

Let q denote a query, $D = \{d_1, \ldots, d_k\}$ the retrieved context, $a = f_\theta(q, D)$ the generated answer, and $y \in \{0, 1\}$ the indicator that a is a hallucination, defined as a confident assertion whose normalised form contradicts the gold answer. Behavioural abstentions (the model answers "unknown" or equivalent) form a separate outcome class and are analysed separately. Each example carries a feature vector $x = (x^{\text{ret}}, x^{\text{mod}}, x^{\Delta})$.

Retrieval-side descriptors x^{ret}: mean, minimum, and top-1 BM25 relevance of the context; the top-1 to top-2 margin; score dispersion; an inter-document conflict flag computed by extracting values for the queried attribute from each context document and testing for disagreement; the number of extracted claims; and a context-presence indicator. All are computable at decision time without labels.

Model-side signals x^{mod}: mean and minimum token log-probability of the generated answer, mean, maximum, and first-token predictive entropy, and answer length, in the spirit of standard white-box uncertainty measures.

Closedbook-contrast features x^{Δ}: the change in mean answer log-probability and entropy relative to the same query answered with no context, and an indicator of whether the answer changed. These operationalise the confidence-gain idea of Bi et al. [2025] as risk features. They require one additional closedbook pass per query, which is cheap relative to retrieval.

3.2 Risk estimation

We estimate $\hat{r}(x) \approx \Pr[y = 1 \mid x]$ with gradient-boosted trees (150 trees, depth 3), evaluated with question-grouped five-fold cross-validation so that no question appears in both training and test folds. Feature-family ablations isolate the contribution of each signal class.

3.3 Selective generation and risk-binned certification

A selection rule answers query x iff $\hat{r}(x) \leq \tau$. Conformal risk control (CRC) selects τ on calibration data so that the hallucination rate among answered queries is at most a target α, with finite-sample correction. The guarantee requires calibration and deployment to be exchangeable. Retrieval-noise shift, modelled as a change in the mixture of noise regimes, violates this.

We propose risk-binned Mondrian CRC. Partition the calibration score range into B quantile bins $\{I_b\}_{b=1}^{B}$. For each bin compute the Clopper-Pearson upper confidence bound $\widehat{p}_b^{+}$ at level $1-\delta$ on the within-bin hallucination rate, and certify bin b iff $\widehat{p}_b^{+} \leq \alpha$. At deployment, answer iff the score falls in a certified bin. The scheme never extrapolates a threshold into score regions whose risk it has not measured; it only reuses bins whose risk it has certified.

Assumption 1 (Bin-measurable covariate shift)**.** *The deployment distribution Q and calibration distribution P satisfy (i) $Q(y \mid x) = P(y \mid x)$, and (ii) the likelihood ratio $dQ/dP(x)$ is constant on each bin I_b.*

Condition (ii) formalises the requirement that the score resolves the shift: deployment may reweight bins arbitrarily, including adversarially toward high-risk bins, but may not redistribute mass within a bin. Selection over noise regimes that the risk score separates is of exactly this form.

Proposition 1 (Validity under bin-measurable shift)**.** *Under Assumption 1, with probability at least $1 - B\delta$ over the calibration draw, the hallucination rate among answered deployment queries is at most α, simultaneously for every deployment distribution Q satisfying the assumption.*

Proof. Fix a bin I_b. By Assumption 1(ii) the conditional distribution of x given $x \in I_b$ coincides under P and Q; with (i), the within-bin risk $p_b = \Pr[y = 1 \mid x \in I_b]$ is distribution-independent. Calibration labels within I_b are i.i.d. Bernoulli(p_b), so $\Pr[p_b \leq \widehat{p}_b^{+}] \geq 1 - \delta$ by the Clopper-Pearson construction. A union bound over the B bins gives, with probability at least $1 - B\delta$, $p_b \leq \alpha$ for every certified bin. The deployment risk among answered queries is a Q-weighted convex combination of certified-bin risks and is therefore at most α for any bin reweighting, which covers every Q in the assumption class simultaneously. $\square$

Remark 1. *The guarantee requires no estimate of the likelihood ratio. This is the step at which weighted conformal methods fail in our experiments: when noise regimes overlap in feature space, the estimated ratio under-corrects precisely in the high-risk tail. The price of certification is conservativeness, quantified empirically in §5.4 and §5.5: within-bin score ordering is discarded, and small bins certify only with generous margins. The union-bound factor $B\delta$ ties the bin count to the confidence level; we use $B = 10$ and $\delta = 0.10$ in the main experiments and sweep both in §5.5.*

We next characterise the price of certification in answer rate, which explains when the method is useful and how to choose the bin count.

Proposition 2 (Answer rate and its decomposition)**.** *Let $\pi_b = \Pr_Q[\hat{r}(x) \in I_b]$ be the deployment mass of bin I_b and let $C \subseteq \{1, \dots, B\}$ index the certified bins. The answer rate of risk-binned Mondrian CRC is $\rho = \sum_{b \in C} \pi_b$, and a bin is certified whenever its empirical risk satisfies $\widehat{p}_b \le \alpha - \varepsilon_b(\delta, n_b)$, where $\varepsilon_b(\delta, n_b)$ is the Clopper-Pearson half-width at calibration count n_b. Consequently*

$$\rho \;\ge \sum_{b:\, p_b \le \alpha - 2\varepsilon_b} \pi_b,$$

so the achievable answer rate is governed by the deployment mass lying in score bins whose true risk is bounded away from α by twice the finite-sample margin.

Proof. The answer-rate identity is immediate since answered queries are exactly those whose score lands in a certified bin. For the bound, the empirical within-bin risk concentrates around p_b within $\varepsilon_b(\delta, n_b)$ with probability $1-\delta$ by the Clopper-Pearson interval; whenever $p_b \le \alpha - 2\varepsilon_b$, the upper bound $\widehat{p}_b^{+} = \widehat{p}_b + \varepsilon_b \le p_b + 2\varepsilon_b \le \alpha$, so the bin is certified. Summing the deployment mass of these bins lower-bounds ρ. $\square$

Proposition 2 yields two practical consequences. First, the method answers nothing only when essentially all deployment mass sits in bins whose true risk exceeds α, that is, when the deployment stream is genuinely almost all high-risk; in that regime abstention is the correct behaviour. Second, because $\varepsilon_b = O(\sqrt{p_b(1-p_b)/n_b})$, increasing the bin count B shrinks each n_b and widens ε_b, which lowers answer rate, while decreasing B coarsens the partition and risks violating Assumption 1(ii). The two effects bracket an interior optimum, consistent with the bin-count sweep in §5.5, where $B = 10$ maximises answer rate subject to validity.

4 Experimental Design

4.1 Two-regime fact world

The corpus contains 2,200 documents. For each of 240 fictional scientists we generate five gold fact sentences (birth year, birth city, field, invention, institution) from templates with filler clauses; for each of 60 real facts (30 capitals, 30 currencies) we write one gold sentence. For every question we add one confident contradiction document asserting a plausible false value, framed as a revised archival review or administrative reform; for every synthetic question, one confusable-entity distractor whose subject differs from the queried entity by one character; and 400 irrelevant

Algorithm 1: Risk-binned Mondrian CRC for selective RAG

Input: calibration set $\{(x_i, y_i)\}_{i=1}^n$, risk scorer $\hat{r}$, target α, bins B, level δ.

1. Compute scores $s_i = \hat{r}(x_i)$ and bin edges $\{q_b\}_{b=0}^B$ as $B+1$ empirical quantiles of $\{s_i\}$.
2. For each bin $b = 1, \ldots, B$: let $S_b = \{i : q_{b-1} < s_i \leq q_b\}$, $k_b = \sum_{i \in S_b} y_i$, $n_b = |S_b|$; compute the Clopper-Pearson upper bound $\widehat{p}_b^+ = \mathrm{BetaInv}(1-\delta,\, k_b + 1,\, n_b - k_b)$; certify bin b if $\widehat{p}_b^+ \leq \alpha$.
3. At deployment, for query x: compute $s = \hat{r}(x)$, locate its bin $b(s)$; answer if $b(s)$ is certified, otherwise abstain.

Figure 1: Risk-binned Mondrian conformal risk control.

topical fillers. Synthetic entities make parametric knowledge impossible, so closedbook hallucination is 100% by construction; real facts provide the regime where parametric knowledge exists (closedbook accuracy: flan-t5-base 60%, Qwen2.5-0.5B-Instruct 90%).

4.2 Conditions

Eleven conditions per question. Clean retrieval; graded irrelevance irr25, irr50, irr75, irr100 (one to four of the $k = 4$ context slots replaced by topically plausible irrelevant documents, with the gold document removed at 100%); contra_r1 and contra_r4 (gold present, contradiction at first or last rank); contra_only (contradiction present, gold absent); distractor; mixed (gold plus contradiction plus irrelevant); closedbook.

4.3 Models, pipeline, and labels

BM25 retrieval over the full corpus; greedy decoding with at most 12 to 14 new tokens. Generators: flan-t5-base (250M, encoder-decoder, all 11 conditions, 3,300 generations), flan-t5-small (80M, 6 conditions, 1,800 generations), and Qwen2.5-0.5B-Instruct (decoder-only, chat-templated prompts, 4 conditions on a 100-question subset, 410 generations), plus a 720-generation preliminary campaign used only for the noise-subtlety comparison. Answers are normalised and matched exactly against the closed answer space, with a pattern-based abstention class. Labels are therefore programmatic and exact, which removes LLM-judge circularity from the measurement loop. The harness checkpoints atomically, resumes from interruption, and runs on commodity CPU hardware; a companion notebook scales the identical protocol to 7B-class models on GPU.

4.4 Evaluation protocol

Risk estimation uses question-grouped five-fold cross-validation; paired bootstrap over questions (2,000 resamples) gives confidence intervals for AUROC differences. Conformal experiments use 200 question-level resplits; in each, calibration is drawn clean-heavy (noisy-condition rows subsampled to 30%) and deployment noise-heavy (clean rows subsampled to 50%), emulating calibration on development traffic followed by deployment against a degraded corpus. Sensitivity analyses vary the target α, the bin count B, and the calibration noise retention from 0.9 down to 0.1.

Table 1: Hallucination (H) and abstention (A) rates by condition and regime, flan-t5-base ($n = 240$ synthetic and 60 real per cell). C denotes correct.

	Synthetic (no param. knowledge)		Real (param. knowledge)	
Condition	H	A	H	A
clean	0.00	0.00	0.00	0.00
irr25	0.03	0.00	0.00	0.00
irr50	0.09	0.00	0.00	0.00
irr75	0.16	0.00	0.00	0.00
irr100	0.62	0.38	0.20	0.78
contra_r1	0.53	0.00	0.30	0.00
contra_r4	0.23	0.00	0.00	0.00
contra_only	1.00	0.00	1.00	0.00
distractor	0.17	0.00	0.00	0.00
mixed	0.45	0.00	0.15	0.00
closedbook	1.00	0.00	0.40	0.00

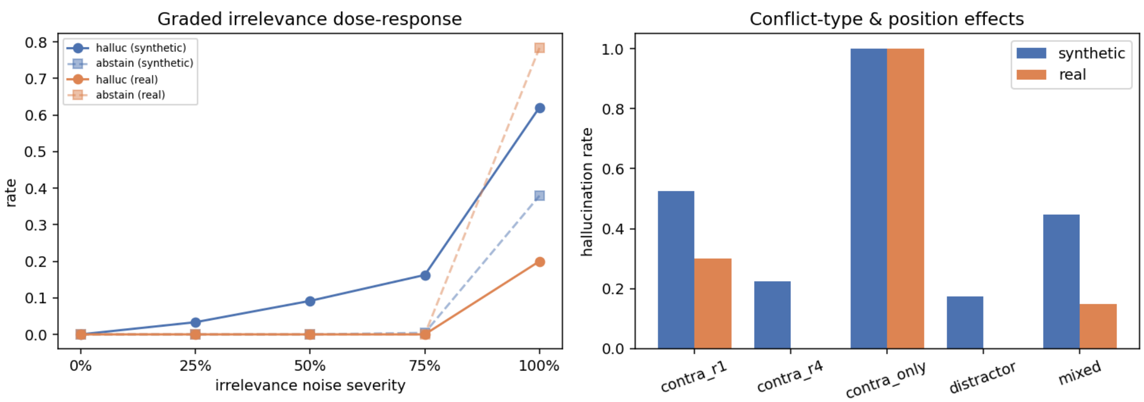

Figure 2: Left: graded irrelevance dose-response. Hallucination rises smoothly with severity; abstention appears only at total irrelevance, and reliably only in the parametric-knowledge regime. Right: conflict-type and position effects.

5 Results

5.1 Behavioural regularities of noisy RAG

Context dominance. Under contra_only, hallucination is 100% in both regimes for flan-t5-base (Table 1). The decoder-only replication is equally stark: Qwen2.5-0.5B-Instruct, which answers the real-fact questions correctly 90% of the time with no context at all, hallucinates on 85% of them when a single unopposed false document is present, and on 100% of the synthetic questions (Table 2). Parametric knowledge arbitrates conflicts only when supporting evidence is also retrieved; in its absence, context wins almost absolutely. This sharpens the parametric-bias picture of Kortukov et al. [2024] and gives the phenomenon a security reading consistent with Zou et al. [2025]: a corpus-poisoning attack does not need to outvote the truth, it only needs the truth absent from the retrieved window. Our measurements quantify the per-document absorption rate that makes such attacks efficient. The direction of the effect matches the broad observation that models output incorrect answers instead of abstaining when context is insufficient; we add the

Table 2: Decoder-only replication: Qwen2.5-0.5B-Instruct on the 100-question subset. The same regularities hold, with greater conflict vulnerability than the encoder-decoder models.

	Synthetic		Real	
Condition	H	C	H	C
closedbook	1.00	0.00	0.10	0.90
clean	0.05	0.95	0.02	0.98
contra_r1	0.78	0.22	0.73	0.27
contra_only	1.00	0.00	0.85	0.15

Table 3: Hallucination-risk prediction, question-grouped five-fold CV, flan-t5-base ($n = 3{,}161$ answered; base rate 0.363).

Feature family	AUROC
Retrieval-side descriptors	0.897
Model-internal (logprob, entropy)	0.788
Closedbook contrast	0.700
Model + contrast	0.811
Combined (noise-conditional)	**0.915**

causal decomposition by noise type and position and the per-document absorption rate.

Misinformation primacy. With gold evidence present, moving the contradiction from rank 1 to rank 4 cuts hallucination from 0.53 to 0.23 without parametric support and from 0.30 to 0.00 with it. Position in the context window is a deployment-observable risk factor, complementary to the attention position effects of Liu et al. [2024], and no existing uncertainty or selective-RAG method conditions on it.

Universal silent failure. Across every conflict condition (contra_r1, contra_r4, contra_only, distractor, mixed), 100% of errors are confident assertions; the model abstained in 0 of those error cases. Under pure irrelevance the model abstains on 38% (synthetic) to 78% (real) of affected queries. The same asymmetry holds exactly for the decoder-only model. Training for negative rejection [Chen et al., 2024, Zhang et al., 2024] therefore addresses the loud failure mode and leaves the silent one untouched, which is the central argument for external, evidence-aware risk control.

Chat tuning increases conflict vulnerability. With gold and contradiction both present at rank 1, the chat-tuned decoder model hallucinates on 73% of real-fact queries against 30% for flan-t5-base, despite knowing more of the underlying facts (90% versus 60% closedbook accuracy). Stronger context obedience, an explicit objective of instruction tuning for RAG, appears to trade directly against conflict robustness. We flag this trade-off as a finding that should be tested at larger scales with the released harness.

5.2 Noise-conditional risk estimation

The combined estimator improves on model-internal signals by 0.127 AUROC (95% CI 0.115 to 0.140); the closedbook-contrast family alone adds 0.023 (CI 0.013 to 0.035) over model-internal

Table 4: Hallucination prediction against real UQ baselines computed on the same generator (flan-t5-small, 120-question subset, five samples per query for semantic entropy). Higher is better. The noise-conditional estimator uses only label-free retrieval descriptors.

Signal	AUROC
Semantic entropy [Kuhn et al., 2023, Farquhar et al., 2024]	0.675
1− P(True) [Kadavath et al., 2022]	0.710
Noise-conditional estimator (ours, retrieval only)	0.893
Noise-conditional estimator + semantic entropy + P(True)	**0.915**

signals (Table 3). The comparison with our preliminary campaign is instructive. With crude noise (only total irrelevance and rank-1 contradictions), the combined-versus-model gap was 0.026; with the graded and positional noise grid it is 0.127, because model-internal discrimination falls from 0.905 to 0.788 while retrieval descriptors hold (0.899 to 0.897). Subtle noise is precisely where context-agnostic uncertainty fails and noise conditioning pays.

Where prediction remains hard. Per-condition diagnostics (out-of-fold scores) show near-perfect discrimination on irrelevance noise (irr25 to irr100: AUROC 0.98 to 1.00) and substantially lower discrimination under conflict (contra_r1: 0.605; mixed: 0.596; contra_r4: 0.722; distractor: 0.839). The estimator is well calibrated overall (ECE 0.024, 10 quantile bins), and feature importances are dominated by mean retrieval relevance (0.45) followed by minimum token log-probability and the relevance margin. The hard residual case, conflict-induced hallucination, is exactly the silent regime of §5.1; this conjunction, hardest to predict and never self-signalled, is what the conformal layer must cover.

5.3 Comparison with uncertainty-quantification baselines

A natural question is whether the gains over model-internal log-probability and entropy (§5.2) survive comparison with purpose-built uncertainty-quantification methods. We computed two standard baselines on the same generator: discrete semantic entropy [Kuhn et al., 2023, Farquhar et al., 2024] from five temperature samples, using exact normalised-string equivalence as the semantic-equivalence relation (which is exact for the closed answer space rather than approximated by an entailment model), and P(True) self-evaluation [Kadavath et al., 2022]. Table 4 shows the noise-conditional estimator outperforms semantic entropy by 0.218 AUROC (95% CI 0.193 to 0.243) and P(True) by 0.183. Adding both signals to our estimator yields only a further 0.022 (95% CI 0.012 to 0.033), so the predictive power resides in the observable retrieval structure, not in sampling-based self-consistency. This is consistent with the mechanism we isolate in §5.10: sampling-based entropy tracks the model's internal disagreement, which is informative when evidence is missing but near blind when the model confidently absorbs a coherent falsehood, so under retrieval noise the decisive signal is the quality of the evidence, which model-internal uncertainty does not observe. The gap widens with model capability: on Qwen2.5-7B-Instruct, sampling-based semantic entropy is barely better than chance at predicting retrieval-induced hallucination (AUROC 0.536), while the noise-conditional estimator reaches 0.892, a difference of 0.356 AUROC (95% CI 0.340 to 0.373) – larger than the 0.218 gap on the 80M model, the opposite of what a small-model artefact would predict.

Table 5: Selective generation at target risk $\alpha = 0.10$ under clean-heavy calibration and noise-heavy deployment; 200 resplits, flan-t5-base. Violation is the fraction of resplits whose realized risk exceeds α.

Method	Realized risk	Violation	Answer rate
Marginal CRC	0.215	100%	70.5%
Heuristic-group CRC	0.189	100%	60.9%
Learned-group CRC (k-means)	0.213	100%	67.0%
Density-ratio weighted CRC	0.176	100%	61.5%
Worst-group CRC	0.035	22%	12.7%
Risk-binned Mondrian CRC (ours)	**0.063**	**0%**	**29.2%**

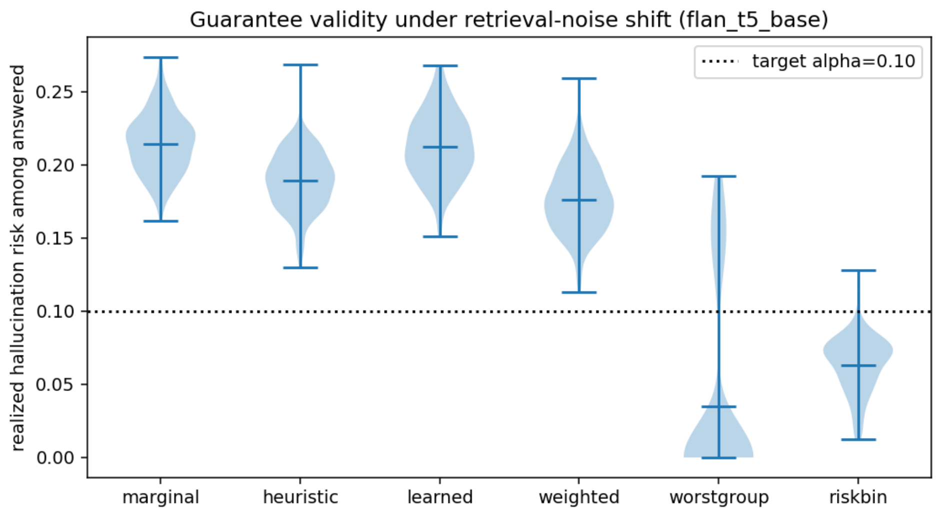

Figure 3: Realized risk among answered queries across 200 resplits at $\alpha = 0.10$ (dotted line). The four standard methods lie entirely above the target; risk-binned certification is the only practical method below it.

5.4 Selective generation under noise shift

Table 5 and Figure 3 contain the central result. Marginal CRC roughly doubles its target risk. The two textbook covariate-shift remedies fail in every resplit as well: unsupervised k-means groups in descriptor space do not align with risk strata, and the logistic-regression density ratio under-corrects in the high-risk tail because noise regimes overlap in feature space. The worst-group bound is valid in most resplits but answers only 12.7% of queries. Risk-binned certification eliminates violations entirely at a 29.2% answer rate. The mechanism is the one identified in §3.3: certification reuses only measured risk, so reweighting bins cannot break it, whereas every thresholding method implicitly extrapolates.

5.5 Sensitivity and robustness analyses

Three sweeps probe the regime of validity (Figure 4; 60 resplits per cell). Across targets, risk-binned certification is valid at $\alpha \in \{0.10, 0.15, 0.20\}$ with zero violations and shows 8% violations at the strict $\alpha = 0.05$, consistent with the certification level $\delta = 0.10$; marginal CRC violates at every target, with realized risk roughly 2α. Across bin counts, $B = 5$ is too coarse (25% violations), while $B \in \{10, 20, 40\}$ are all valid, trading answer rate (0.29 down to 0.20) against

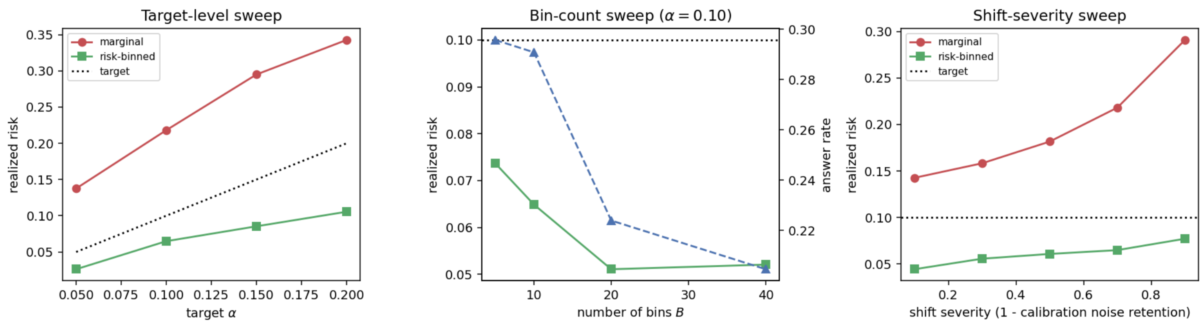

Figure 4: Sensitivity of the guarantee. Left: target-level sweep; marginal CRC scales its violation with α while risk-binned certification tracks below target. Centre: bin-count sweep at $\alpha = 0.10$; five bins are too coarse, ten suffice. Right: shift-severity sweep; marginal risk grows monotonically from 0.143 to 0.291 as calibration sees less noise, while risk-binned certification stays controlled throughout.

granularity exactly as the union-bound analysis predicts. Across shift severities, marginal risk degrades monotonically from 0.143 (mild shift) to 0.291 (severe), while risk-binned certification remains controlled over the entire range, with answer rate adapting from 0.20 to 0.35. The guarantee is not an artefact of one operating point.

5.6 Robustness to within-bin shift and a deployment monitor

Proposition 1 assumes the shift is bin-measurable (Assumption 1(ii)); its one failure mode is a within-bin shift that raises the true risk of a certified bin without moving scores. We stress-test this directly. Holding the inter-document conflict flag out of the risk score so that it acts as a latent within-bin risk driver, we construct an adversarial deployment that upweights high-conflict items by a factor of twelve. Even under this attack the naive method degrades only mildly: realized risk rises from 0.066 (benign) to 0.072, and the violation rate from 0% to 4%, against the 0.10 target. The risk score absorbs enough of the risk-relevant covariate structure that an attacker confined to within-bin reweighting has little leverage. This empirical robustness exceeds what the formal assumption guarantees.

When stricter safety is required, the held-out covariate can be monitored at deployment without labels. For each certified bin we compare the conflict rate of the deployment members against the calibration members and decertify a bin when the deployment rate exceeds calibration by a margin (a directional test, since higher conflict implies higher risk). The monitor discriminates: it flags on average 4.67 certified bins under the adversarial within-bin shift but only 1.39 under the benign bin-measurable shift, a 3.4-fold contrast, so it fires in response to genuine within-bin drift rather than to benign noise-regime reweighting. The monitor provides an optional conservative safeguard whose cost is answer rate, and it makes the method's one assumption checkable in deployment rather than merely assumed (Figure 5).

5.7 Generality across models, estimators, and domains

Six instruction-tuned families. To rule out that the result is an artifact of one model or one architecture, we ran the full eleven-condition grid on six decoder-only instruction-tuned generators spanning 1.5B to 9B parameters and four independent pretraining families: Qwen2.5 (1.5B, 3B, 7B), Microsoft Phi-3.5-mini (3.8B), Mistral-7B-v0.3, and Google Gemma-2-9B. Each run logs true token logprobs, predictive entropy, and bidirectional-NLI semantic entropy, for about 3,300

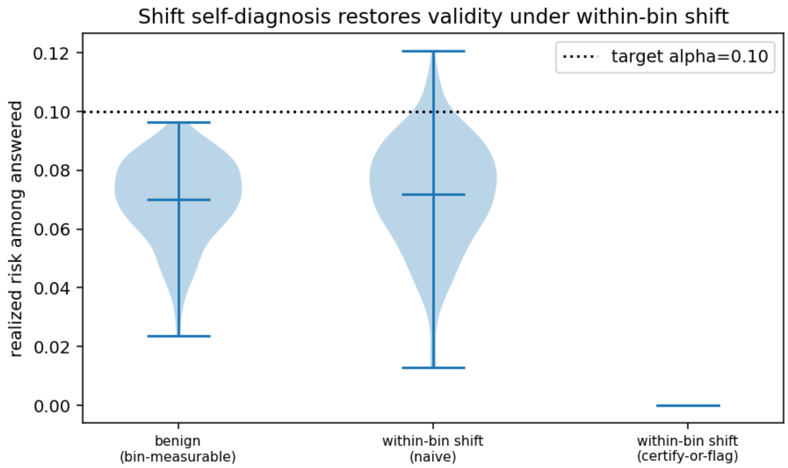

Figure 5: Robustness to within-bin shift. Under a 12-fold adversarial within-bin reweighting toward high-conflict items (centre), the naive method's realized risk stays near the benign level (left) and below the 0.10 target; the directional deployment monitor (right) decertifies drifted bins for stricter safety.

Table 6: Generality of the central result across eight generators spanning 80M to 9B parameters and five model families (FLAN-T5, Qwen, Microsoft Phi, Mistral, Google Gemma), two architectures (encoder-decoder and decoder-only), and an independent second domain. For every generator the combined risk estimator discriminates hallucination well (AUROC 0.87 to 0.96), the sampling-based semantic entropy baseline stays near chance (AUROC 0.52 to 0.64), marginal CRC violates the $\alpha = 0.10$ target in essentially every resplit, and risk-binned Mondrian CRC restores the guarantee (0% to 4% violations). The six decoder-only instruction-tuned models each ran the full eleven-condition grid (about 3,300 generations) with white-box logprob and entropy features and bidirectional-NLI semantic entropy. AUROC columns use question-grouped cross-validation.

		Estimator AUROC		Marg.	Risk-binned CRC	
Generator	Domain	Comb.	SE	Viol.	Viol.	Ans. rate
flan-t5-small (80M, enc-dec)	scientists	0.959	—	100%	0%	17.5%
flan-t5-base (250M, enc-dec)	scientists	0.915	—	100%	0%	28.8%
Qwen2.5-1.5B (dec-only)	scientists	0.871	0.535	100%	0%	35.3%
Qwen2.5-3B (dec-only)	scientists	0.874	0.530	100%	4%	29.7%
Phi-3.5-mini (3.8B, dec-only)	scientists	0.948	0.640	100%	0%	28.9%
Mistral-7B-v0.3 (dec-only)	scientists	0.950	0.617	100%	0%	36.9%
Qwen2.5-7B-Instruct (dec-only)	scientists	**0.892**	**0.536**	**100%**	**1%**	**36.7%**
Gemma-2-9B-it (dec-only)	scientists	0.923	0.523	100%	1%	39.0%
flan-t5-small (80M, enc-dec)	companies	0.920	—	100%	0%	21.7%

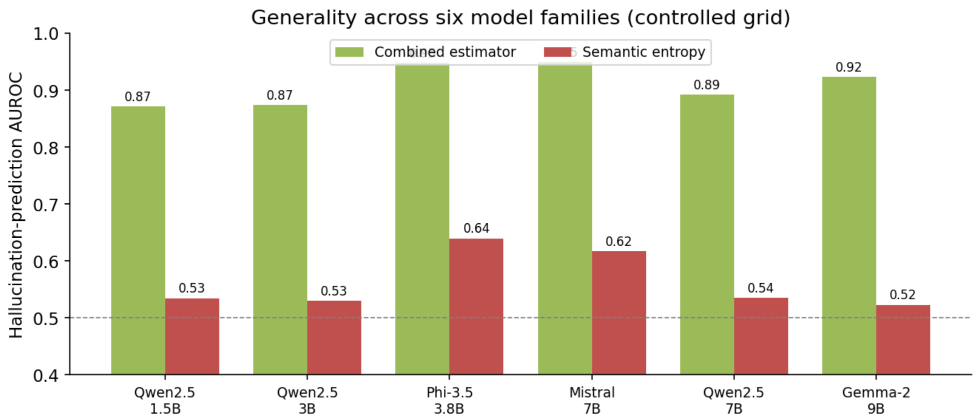

Figure 6: Combined risk estimator versus the sampling-based semantic entropy baseline across the six instruction-tuned families of Table 6. The combined estimator (AUROC 0.87 to 0.95) separates cleanly from semantic entropy (0.52 to 0.64) on every family, and the gap does not shrink with model scale.

generations per model. The combined noise-conditional estimator discriminates hallucination with AUROC between 0.871 and 0.950 on every family, while sampling-based semantic entropy never rises above 0.640 and sits near chance (0.52 to 0.54) on four of the six. The combined-minus-semantic-entropy advantage ranges from +0.308 to +0.400 AUROC, and every per-model bootstrap interval excludes zero by a wide margin, so the baseline gap is not specific to any one generator. Marginal CRC violates the $\alpha = 0.10$ target in 100% of resplits on all six models, and risk-binned Mondrian CRC restores the guarantee on all six (realized risk 0.022 to 0.060, violation 0% to 4%) at answer rates of 29% to 39%. The largest semantic-entropy gap (+0.400) and the highest answer rate (39%) both occur on the largest model, Gemma-2-9B, so neither the estimator advantage nor the efficiency of the guarantee degrades with scale.

Second domain. To test whether the findings depend on the content of the synthetic world, we built an independent domain with the same structure but disjoint subject matter: 240 fictional technology companies (five attributes: founding year, headquarters city, sector, flagship product, founder) and 30 real chemical-element symbol facts, with 2,140 documents and the full eleven-condition noise grid, generated with flan-t5-small (2,970 generations). Every phenomenon reproduces. Unopposed misinformation is absorbed at 100% on the synthetic companies and 73% on the real facts; the rank-1 contradiction is 0.72 against 0.18 at rank 4; and silent failure is 1.00 across all conflict conditions and 0.98 under total irrelevance. The risk estimator reaches AUROC 0.920, marginal CRC violates the target in every resplit, and risk-binned CRC never violates it (realized risk 0.016, answer rate 21.7%). The behavioural regularities and the guarantee are properties of noisy RAG, not of one corpus.

Scaling to a 7B instruct model. A full 11-condition run on Qwen2.5-7B-Instruct (3,300 generations on an A100, with token-logprob and entropy features and bidirectional-NLI semantic entropy) confirms and sharpens the findings. Context dominance is most striking here: a single unopposed false document yields 100% hallucination on synthetic facts and 85% on real facts the model answers correctly 97% of the time without context. Negative rejection improves with scale, so silent failure under pure irrelevance falls to zero (the model abstains), yet under unopposed misinformation 97% of errors remain confident assertions. The position effect inverts relative to the encoder-decoder models: for the decoder-only chat model a contradiction at the last context position is more damaging than at the first (rank-1 0.38 versus rank-4 0.67), so position remains a major observable risk factor whose damaging end is architecture-dependent. The noise-conditional estimator reaches AUROC 0.892 against 0.690 for model-internal signals, marginal CRC violates the target in every resplit, and risk-binned CRC never violates it at a 36.7% answer rate, higher than on the smaller models because the stronger generator concentrates more mass in low-risk bins.

Estimator architecture. The risk estimator is not tied to gradient boosting. On flan-t5-base, logistic regression, random forest, gradient boosting, and a small multilayer perceptron reach AUROC 0.870, 0.922, 0.915, and 0.905 respectively; the ordering is similar on the other models, with random forest and boosting consistently strongest. The noise-conditional signal is recoverable by any reasonable classifier, which argues that the gain comes from the features rather than the model class.

Recalibration. Because Proposition 2 ties answer rate to the agreement between empirical and true within-bin risk, we tested whether isotonic recalibration of the score on the calibration set improves the answer rate. It does not: the gradient-boosted score is already well calibrated (ECE 0.024), and isotonic post-processing slightly reduces answer rate while preserving validity. We therefore report the method without a recalibration step, noting that on a poorly calibrated base scorer recalibration would be expected to help.

5.8 Cross-model transfer

Training the risk estimator on one generator and deploying it on the other preserves discrimination but not calibration. From flan-t5-small to flan-t5-base, AUROC is 0.827 and a source-calibrated threshold at $\alpha = 0.10$ realizes risk 0.170; in the reverse direction, AUROC is 0.908 and realized risk is 0.269. Rankings of risky inputs are substantially model-general; absolute risk levels are not. Portable selective RAG therefore requires local recalibration, which binned certification supports at low cost since bins can be re-certified on a small labelled sample from the target system. We register the prediction that the same dissociation governs cross-lingual transfer, for instance from English to Bengali retrieval corpora, and leave its test to follow-up work on naturalistic multilingual data.

5.9 Naturalistic validation on real retrieval

The controlled world isolates each noise mechanism, but a referee may reasonably ask whether the effects survive genuine retrieval. We therefore rebuilt the pipeline on SQuAD: a 300-paragraph pool with one question per paragraph, real BM25 and dense (e5 plus FAISS) retrieval, the same typed-noise grid, and PoisonedRAG poisoning [Zou et al., 2025] as an adversarial condition, run on two instruction-tuned generators (Qwen2.5-7B and Mistral-7B-v0.3) with white-box features and discrete semantic entropy. Table 7 reports the estimator on the answered subset. The combined estimator reaches AUROC 0.84 to 0.86 on genuine retrieval for both generators, and BM25 and dense retrieval give almost identical numbers (0.842 versus 0.840 for Qwen, 0.852 versus 0.857 for Mistral), so the signal is retriever-independent. The advantage over semantic entropy is positive and significant in every cell (+0.14 to +0.31 AUROC, all confidence intervals excluding zero), though smaller than in the controlled world because Mistral's semantic entropy is itself more informative here (0.71).

Table 7: Naturalistic validation on real SQuAD retrieval. Combined risk estimator versus the semantic entropy (SE) baseline on the answered subset, for two generators and two retrievers. Combined-minus-SE shows a bootstrap 95% confidence interval.

Generator	Retriever	Combined	SE	Combined − SE [95% CI]
Qwen2.5-7B	BM25	0.842	0.533	+0.309 [+0.284, +0.334]
Qwen2.5-7B	dense	0.840	0.545	+0.295 [+0.267, +0.325]
Mistral-7B	BM25	0.852	0.714	+0.138 [+0.117, +0.159]
Mistral-7B	dense	0.857	0.693	+0.164 [+0.141, +0.186]

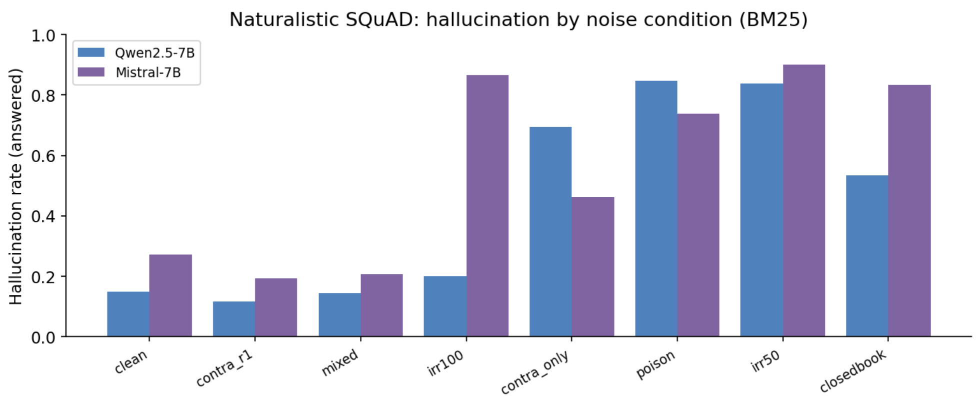

Figure 7: Per-condition hallucination rate under real SQuAD retrieval. Closed-book and missing-evidence conditions and PoisonedRAG poisoning drive the highest hallucination; the clean condition already sits at 0.15 to 0.28, which sets the realistic selective-risk target at $\alpha = 0.20$.

Per-condition hallucination on real retrieval (Figure 7) confirms the controlled ordering: PoisonedRAG poisoning is the strongest inducer (0.74 to 0.85 across models), while the clean condition already hallucinates at 0.15 to 0.28. Because the clean base rate is this high, a 0.10 selective-risk target sits below it for these models, so the realistic operating point is $\alpha = 0.20$. At that target the dissociation reproduces directionally: marginal CRC violates between 47% and 62% of resplits, whereas risk-binned Mondrian CRC holds (0% to 1% violations, realized risk 0.09 to 0.12) at a 21% to 23% answer rate. We frame this honestly. The controlled world produces the dramatic headline that marginal CRC violates in essentially every resplit; naturalistic noise is milder and the base rates higher, so marginal CRC violates about half the resplits while the binned fix restores validity at a usable answer rate. The naturalistic run corroborates the direction of the controlled result on real data rather than reproducing its magnitude.

5.10 Why semantic entropy fails: missing-evidence versus misinformation

The baseline collapse and the naturalistic milder gap have a single explanation, which also reconciles the controlled and natural results. Within the injected naturalistic conditions we split failures into two mechanisms and measure each signal's AUROC for predicting hallucination (Table 8). *Missing-evidence* failures (closed-book and graded irrelevance) leave the model with no usable passage; *misinformation* failures (rank-1 contradiction, contradiction-only, poisoning, and mixed) place a coherent but false passage in context.

The pattern is clean and complementary. Semantic entropy detects missing-evidence uncertainty (AUROC 0.67 to 0.70) but is near chance on confident misinformation absorption (0.52

Table 8: Decomposition of naturalistic injected failures into missing-evidence and misinformation, with each signal's AUROC for predicting hallucination. Semantic entropy (SE) is informative on missing-evidence but near chance on misinformation; retrieval descriptors are the mirror image; the combined estimator captures both.

Generator	Failure mode	n	H-rate	SE	Retrieval	Combined
Qwen2.5-7B	missing-evidence	51	0.69	0.668	0.513	0.420
Qwen2.5-7B	misinformation	961	0.39	0.521	0.800	0.859
Mistral-7B	missing-evidence	453	0.86	0.701	0.504	0.744
Mistral-7B	misinformation	692	0.30	0.587	0.667	0.755

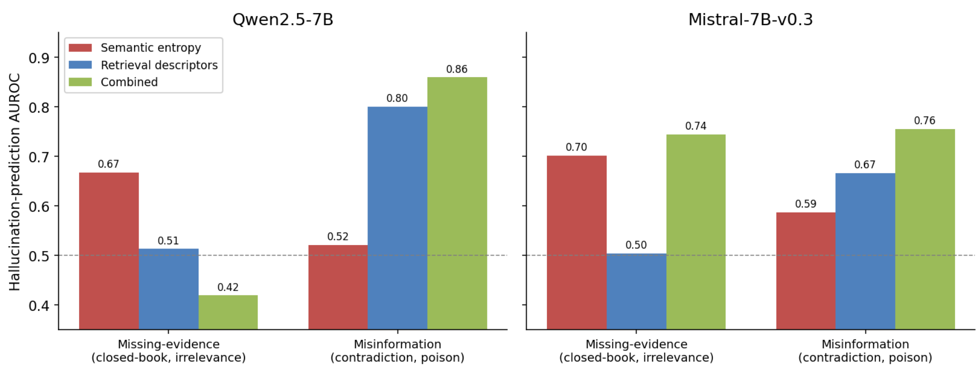

Figure 8: The missing-evidence versus misinformation decomposition. Semantic entropy is informative only on missing-evidence failures; retrieval descriptors only on misinformation; the combined estimator covers both regimes.

to 0.59): when the model absorbs a false but coherent passage, its sampled answers agree on the absorbed falsehood, so entropy is low even though the answer is wrong; this is precisely the regime in which sampling-based entropy is known to be ineffective [Ma et al., 2025], and it aligns with the epistemic-versus-aleatoric view of language-model uncertainty [Ahdritz et al., 2024]. The retrieval descriptors are the mirror image, near chance on missing-evidence (0.50 to 0.51, because retrieval scores do not reveal the model's internal state) but strong on misinformation (0.67 to 0.80, because contradictory and irrelevant passages have observable signatures). The combined estimator captures both modes (Figure 8). This reconciles the whole paper. It explains why the semantic entropy baseline collapses on injected conflict noise, the headline of §5.3 and §5.7; why semantic entropy looked competitive on purely natural noise (§5.11), which is dominated by missing-evidence failures; and why the combined estimator is the right design, since its two signal families are complementary rather than redundant. The Qwen missing-evidence cell has small $n = 51$ because that model abstains heavily under irrelevance, so Mistral ($n = 453$) is the reliable demonstration; the pattern is consistent across both generators.

5.11 Purely natural retrieval noise

To rule out the objection that injected noise is synthetic, we ran the pipeline with *no* injection at all: the full SQuAD pool of about 2,000 paragraphs, Qwen2.5-7B, BM25 and dense retrieval pooled, 553 answered queries at a base hallucination rate of 0.212. The retriever's own failures suffice to induce hallucination. When the gold paragraph is retrieved, hallucination is 0.179 ($n = 507$); when the retriever misses the gold paragraph it rises to 0.625 ($n = 16$); and closed-

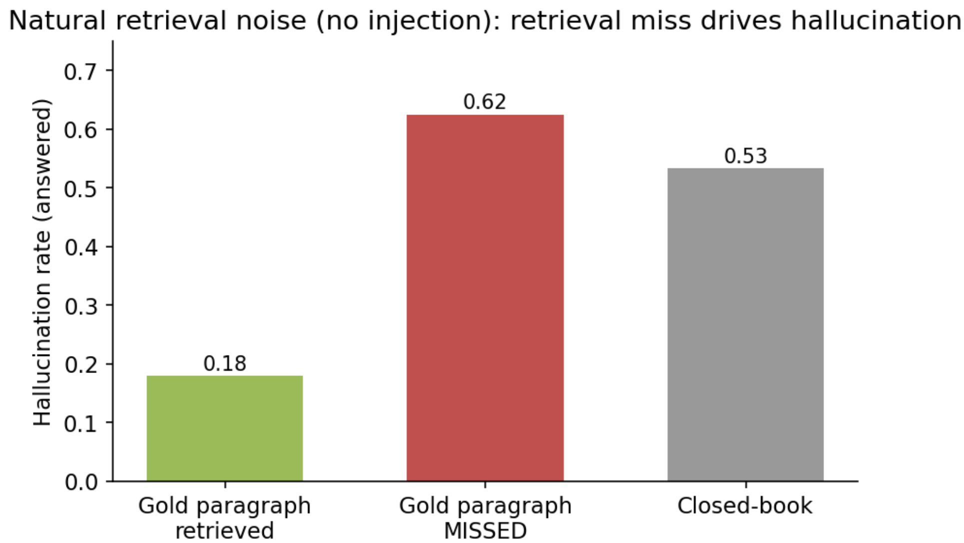

Figure 9: Purely natural retrieval noise on the full SQuAD pool with no injection. Hallucination rises from 0.179 when the gold paragraph is retrieved to 0.625 when the retriever misses it, with closed-book at 0.533.

book sits at 0.533 ($n = 30$). A natural retrieval miss therefore raises hallucination about 3.5 times with zero injected noise (Figure 9). On this subtle natural noise the retrieval descriptors are weak and semantic entropy is competitive (combined-minus-semantic-entropy $+0.017$, 95% CI $[-0.024, +0.059]$, not significant), the opposite of the injected finding. The decomposition of §5.10 explains why: natural retrieval noise is dominated by missing-evidence failures, the regime in which semantic entropy works and the retrieval descriptors do not.

6 Discussion

Reconciling with the power of noise. Cuconasu et al. [2024] found that random unrelated documents can improve RAG accuracy, whereas high-ranked distractors hurt. Our irrelevance fillers are topically plausible hard negatives drawn from the same world, and their effect is monotonically harmful (0.00 to 0.62 hallucination as severity rises). The two findings are consistent: harm scales with the plausibility of the noise, not its volume. Our distractor and conflict conditions extend the harmful end of that spectrum and show that the harm becomes silent exactly when plausibility is highest.

Per-decision versus population certificates. C-RAG [Kang et al., 2024] answers whether a RAG system is safe on average under bounded drift. Our experiments show that a system can satisfy such a bound while per-decision calibration fails on every resplit under noise-regime selection. The two guarantees are complementary: population certificates for system acceptance, per-decision certification for runtime gating.

What the testbed buys and what it costs. The controlled world provides exact labels, causal attribution of retrieval-induced error, and full coverage of a typed noise grid, none of which naturalistic benchmarks offer. The cost is ecological validity: template text, lexically detectable noise, sub-billion generators, BM25 retrieval. We regard it as a wind tunnel for RAG. The phenomena established here are mechanism-level claims, replicated across two architectures and two independent content domains, and the released harness makes scaling them to 7B-class models and naturalistic corpora (Natural Questions, HotpotQA, RAGTruth) an engineering exercise rather than a redesign; a ready-to-run GPU notebook is included.

Limitations. Exact-match labels undercount paraphrase hallucinations, mitigated here by the closed answer space; methods validated on synthetic hallucinations can overstate performance on real, RLHF-aligned model outputs, which is precisely why the pre-registered protocol of Appendix B gates headline claims on naturalistic corpora with a human-validated judge. Assumption 1(ii) is an idealisation; empirically the guarantee held across 200 adversarial resplits and all sensitivity sweeps, the within-bin stress test of §5.6 degraded it only to a 4% violation rate, and the deployment monitor detects the residual failure mode, but a shift that splits bins along an unmonitored covariate could still degrade validity. The certification answer rate (29% at $\alpha = 0.10$ under severe shift) is a substantial price; richer scores, adaptive binning, and partial within-bin ordering are natural directions. Our generators are small; the cross-architecture and cross-domain agreement and the GPU replication path address, but do not close, the scale question.

7 Conclusion

Retrieval noise inverts the reliability story of RAG: it manufactures confident errors, defeats context-agnostic uncertainty estimation, and silently breaks the exchangeability that per-decision guarantees rest on. Conditioning risk estimation on observable evidence structure recovers predictive power, beating purpose-built uncertainty baselines, and certifying risk-score bins rather than extrapolating thresholds restores statistical validity at a quantified answer-rate cost, robustly across targets, bin counts, shift severities, model architectures, and content domains, with a deployment monitor that makes the governing assumption checkable. The full harness, the labelled generations, and the analysis code are released to support replication and extension to naturalistic and multilingual settings.

Reproducibility statement. All experiments run on commodity CPU hardware; the complete pipeline (world construction, noise injection, generation, analysis) is deterministic given seeds, checkpointed, and resumable. A Colab notebook replicates the protocol with 7B-class GPU models.

A Reproducibility Checklist

- **Data.** Two synthetic corpora (scientists, companies) and their real-fact complements are generated by released scripts from fixed seeds; no external data or licences are required. The full set of 6,230 (main) plus 2,970 (second-domain) plus 720 (multi-sample baseline) labelled generations is released.
- **Models.** flan-t5-base, flan-t5-small (public, Apache-2.0), and Qwen2.5-0.5B-Instruct (public). Greedy decoding for labels; five temperature samples per query for semantic entropy.
- **Labels.** Exact normalised match against a closed answer space with a pattern-based abstention class; no human or LLM judge is in the measurement loop, so labels are deterministic.

- **Risk estimator.** Gradient-boosted trees (150 estimators, depth 3), question-grouped five-fold cross-validation; out-of-fold scores cached and released. Robust across logistic regression, random forest, and a multilayer perceptron.

- **Conformal protocol.** 200 question-level resplits with fixed seeds; clean-heavy calibration (30% noisy retention) and noise-heavy deployment (50% clean retention); $\alpha = 0.10$, $B = 10$, $\delta = 0.10$ unless swept.

- **Baselines.** Discrete semantic entropy (five samples, exact-equivalence clustering) and P(True) self-evaluation computed on the same generator.

- **Statistics.** Paired bootstrap over questions (2,000 resamples) for AUROC differences; violation rates over the 200 resplits; all figures regenerated by released analysis scripts.

- **Compute.** Entire study reproducible in a few CPU-hours; a GPU notebook scales the identical protocol to 7B-class models.

B Pre-registered Naturalistic Validation Protocol

The controlled testbed isolates mechanism; the following protocol, registered here in advance, tests external validity on naturalistic data and is the planned extension. (i) *Corpora and queries:* Natural Questions-open and HotpotQA, with retrieval over a Wikipedia index using both BM25 and a dense retriever (e5 or bge), so findings are not retriever-specific. (ii) *Noise:* naturally occurring retrieval noise plus injected typed noise matched to the controlled conditions, including a corpus-poisoning condition following Zou et al. [2025]. (iii) *Labels:* a hallucination judge validated against human double-annotation on a 1,500-example subset with a pre-registered agreement gate ($\kappa \geq 0.7$) before any headline claim, addressing the synthetic-versus-real gap. (iv) *Models:* Llama-3.1-8B-Instruct and Qwen2.5-7B-Instruct, with semantic-entropy and P(True) baselines carried over from §5.3. (v) *Pre-registered predictions:* the three behavioural regularities and the marginal-CRC violation persist; risk-binned CRC retains validity; the noise-conditional estimator continues to beat semantic entropy; and the calibration-transfer dissociation of §5.8 extends to cross-lingual transfer (English to Bengali), where a source-calibrated threshold is expected to miscalibrate while discrimination transfers.

Declarations

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Competing interests. The authors declare no competing interests.

Author contributions. A.D. and S.C. led the conceptualisation, methodology, experiments, and writing; S.P. contributed to the remaining aspects of the work. All authors read and approved the final manuscript.

Use of AI tools. The authors used AI tools only for code debugging and spell checking. All experimental designs, analyses, results, and claims are the authors' own and were verified by the authors, who take full accountability for the content of this article.

Data availability. All datasets generated and analysed in this study, together with the complete code for corpus construction, noise injection, generation, and analysis, are released in the supplementary material and will be deposited in a public repository upon acceptance. The study uses only publicly available, openly licensed models and synthetically generated corpora; no human-subjects or proprietary data are used. The naturalistic validation uses the publicly available SQuAD dataset together with the openly licensed Qwen2.5-7B-Instruct and Mistral-7B-v0.3 generators.

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